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End Semester Examination 2024

Name of the Course: B. Tech, EE

Semester: 3rd

Name of the Paper – Electromagnetic Fields

Paper Code: TEE 304

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	What is the physical significance of the gradient, divergence, and curl of a vector or scalar field, and how is it represented geometrically?	CO1
(b)	Convert the point $(r, \theta, z) = (4, \pi/3, 5)$ in cylindrical coordinates to Cartesian and Spherical coordinate systems.	
(c)	If a point in spherical coordinate system is located as (r, θ, Φ) then find unit vector in the direction of increasing r, θ , and Φ .	
Q2		
(a)	Discuss Gauss's law of electrostatics and magnetism	CO2/CO3
(b)	Using Gauss's law, derive the electric field intensity at a point due to infinite sheet with charge per area equal to σ	
(c)	Four-point charges of magnitude 1 nC, 2 nC, 3nC, and 4nC are situated at the vertex of square of side length 1 c.m. Find electric field at the center of the square.	
Q3		CO3/CO4
(a)	Discuss displacement current and derive its expression using Ampere Circuital law	
(b)	What is the continuity equation in the context of current flow, and how does it express the principle of charge conservation? Also, derive continuity equation.	
(c)	If potential (V) at any point is given as (i) $V(x, y, z) = 5x^3y^2z$ (ii) $V(\rho, \phi, z) = z \cos(\phi)/\rho$ then find charge per unit volume at $(-3, 1, 2)$ and $(3, 30^\circ, 2)$.	
Q4		CO5
(a)	Using Biot and Savart law, derive the magnetic field at an axial distance of 'x' of center of ring. Assume radius of ring = 'a', number of turn = N, and current flowing through the ring = I.	
(b)	Find the Laplacian of the scalar field (i) $V = e^{-z} \sin(2x) \cos(hy)$ (ii) $V = \rho^2 z \cos(2\phi)$	
(c)	Discuss Faraday law of electromagnetic induction. Derive Maxwell equation using Faraday law of electromagnetic induction	
Q5		CO6
(a)	Discuss the differential and integral form of Maxwell's equations.	
(b)	Derive the wave equations for conducting medium.	
(c)	A conducting bar can slide freely over two conducting rails as shown in figure. Calculate the induced voltage in the bar (a) If the bar is stationed at $y = 8$ cm and $\mathbf{B} = 4\cos(10^6 t)\mathbf{a}_z$ mWb/m ² (b) If the bar slides at a velocity $\mathbf{u} = 20\mathbf{a}_y$ m/s and $\mathbf{B} = 4\mathbf{a}_z$ mWb/m ²	

